

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 11: Flight Hazards I: Icing, Turbulence & Mountain Waves

Syllabus Source: CAE Oxford (pp. 359-398) | Pass Target: >= 90% AviationExam

1. AIRFRAME STRUCTURAL ICING PHYSICS

Structural icing occurs when an aircraft flies through visible liquid moisture (clouds/rain) at temperatures below 0°C:

Icing Type	Cloud Type & Temperature Range	Physical Structure & Aerodynamic Severity
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Rime Ice (Rough / Opaque)	Stratiform clouds (St, Sc, As) at -10°C to -20°C. Small supercooled droplets.	Freezes instantly upon impact without spreading. Traps air, milky white, brittle. Alters aerofoil camber, increases drag.
Clear Ice (Glaze / Transparent)	Cumuliform clouds (Cu, Cb) and FZRA at 0°C to -10°C. Large supercooled droplets.	Freezes slowly, spreading aft beyond de-icing boots. Heavy, hard, tenacious. SEVEREST ICING HAZARD in aviation!
Mixed Ice	Mixed clouds at -10°C to -15°C. Combination of small and large droplets.	Rough, hard, difficult to shed with de-icing boots.
Hoar Frost	Forms on parked aircraft overnight, or descending rapidly into warm, moist air.	Crystalline frost. Degrades boundary layer: INCREASES STALL SPEED by up to 33% and reduces lift by 30%!

2. MOUNTAIN WAVES (LEE WAVES)

Mountain waves develop on the leeward side of mountain ridges under specific conditions:

- Necessary Conditions: 1. Wind direction within 30° of perpendicular to ridge; 2. Wind speed at crest level >= 20 knots and increasing with altitude; 3. Very stable air layer sandwiched between less stable layers above and below.
- Wave Cloud Features: Cap cloud (Föhn wall) over crest; Lenticular clouds (Alto cumulus lenticularis) stationary at wave crests; Rotor clouds (Roll clouds) formed beneath wave crests with VIOLENT, DESTRUCTIVE TURBULENCE!
- Wave Wavelength Rule of Thumb: Wavelength (NM) ~ 0.5 x Wind speed across ridge (knots) (e.g. 40 kt wind -> ~20 NM wavelength).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Rime Ice vs Clear Ice & Mountain

Structural Icing Rules: - Clear Ice: 0 C to -10 C in CB/Cu with LARGE supercooled droplets. Transparent, heavy, spreads backward beyond de-icing boots. Extremely dangerous! - Rime Ice: -10 C to -20 C in St/Sc with SMALL droplets. Freezes instantly on impact, trapping air. Milky, opaque, brittle. Mountain Wave Survival Rules: - Conditions: Wind perpendicular to ridge ≥ 25 kt, increasing with height, stable layer aloft. - Hazards: Rotor clouds below crests contain EXTREME turbulence; downdrafts exceed 3,000 to 5,000 ft/min on lee side! - Rule: Always cross ridges at a 45-degree angle (never 90 degrees) to allow an immediate turn away toward low ground!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Mountain Wave Rotor & Descent Hazards**

SCENARIO (Mountain Flight Safety Drill): A twin-engine aircraft is flying across the Pyrenees: - Mountain ridge height = 8,000 ft MSL - Wind at crest = 010° at 40 knots (directly perpendicular to ridge) - The leeward side exhibits Altocumulus lenticularis clouds at 12,000 ft and ragged roll clouds at 5,000 ft AGL QUESTION: What is the estimated distance between wave crests, and what lethal hazard lies inside the roll clouds? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Mountain Wave Wavelength: - Formula: Wavelength (NM) $\sim 0.5 \times$ Wind Speed (knots) - Wavelength = 0.5×40 kt = 20 Nautical Miles! - (Successive downdrafts and updrafts will repeat every 20 NM downwind of the mountain!). Step 2: Identify the Hazards in the Roll Cloud (Rotor Zone): - The roll cloud marks a closed atmospheric vortex (ROTOR). - Air rotates violently about a horizontal axis. - Updrafts and downdrafts exceed 3,000 to 5,000 ft/min. - Flying into a rotor zone can cause structural overload, control loss, and violent barometric altimeter fluctuations! Step 3: Pilot Action: - Never attempt to penetrate roll clouds! Fly at least 3,000 to 5,000 ft ABOVE the mountain crests, and approach ridges at a 45° angle to permit a retreat turn into the valley.

[!] EXAM TRAP (AVIATIONEXAM) - Hoar Frost Lift-Off Trap

- Even a paper-thin layer of hoar frost on the upper wing surface ruins the laminar boundary layer! - It INCREASES STALL SPEED BY UP TO 33%, and REDUCES MAXIMUM LIFT BY 30%! - Take-off with frost on critical surfaces is STRICTLY ILLEGAL (Clean Aircraft Concept).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under what cloud and temperature conditions is SEVERE CLEAR ICE most likely to form on an aircraft?**

- [A]: Thin Cirrostratus at -40°C
- [B]: CUMULIFORM clouds (Cu, Cb) with large supercooled droplets at temperatures between 0°C and -10°C
- [C]: Fog at $+5^\circ\text{C}$
- [D]: Stratocumulus at -30°C

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Large supercooled droplets have high kinetic energy; upon impact, latent heat of fusion prevents instant freezing, allowing water to flow back and form solid, dense clear ice.

Q2: What atmospheric conditions are ESSENTIAL for the formation of marked MOUNTAIN WAVES?

- [A]: Calm winds and thick fog
- [B]: Wind blowing nearly PERPENDICULAR to the ridge at ≥ 20 knots, increasing with altitude, and a STABLE atmospheric layer over the crest
- [C]: Intense thunderstorm activity
- [D]: Strong tailwind on landing

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The stable layer acts as an elastic spring: air displaced over the ridge oscillates downstream, creating stationary gravity waves.

Q3: How does a small accumulation of frost on the wing surface affect aerodynamic performance during take-off?

- [A]: Decreases stall speed
- [B]: Increases drag and DECREASES MAXIMUM LIFT significantly (can cause premature stall on rotation)
- [C]: Has zero effect
- [D]: Increases ground roll friction only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Frost acts like coarse sandpaper, tripping laminar airflow into turbulent flow early, dramatically reducing maximum lift coefficient Cl_{max} .